



Prince Sultan University
**College of Humanities and
Sciences Dept. of Math and
Sciences**
Second Term 2025- 2026

INSTITUTIONAL COURSE SYLLABUS

Course Code: Math221	Course Title: Numerical Analysis
Course Instructor: Salma Haque	Email: shaque@psu.edu.sa
Credit Hours: 3 hrs	Course Location: Female Campus
Scheduled Office Hours: SMTW: 10am-11am and Thurs 2pm-3pm	
Office Location: R207 Building -101	
Co-Requisite:	Pre-Requisite: Math111
On Campus or Online -On Campus	

Mission Statement

PSU Mission:

Prince Sultan University aims to provide the Middle East with quality education to the highest international standards. In its efforts towards successful and responsible life-long learning, PSU integrates modern technology, pedagogy, and human values for the advancement of scientific research, productivity, and leadership towards a more meaningful role in society.

PSU is committed to effective management of institutional resources to optimize its multiple roles as a catalyst for new learning opportunities, national and international partnerships, continuous studies, professional growth, community service, and diversity in educational horizons for the good of humanity.

CHS Mission:

The College of Humanities and Sciences aspires to provide students with a high-quality education in the fields of humanities and sciences to advance their scientific knowledge, enhance their cultural awareness, broaden their horizons, and create a balance in student life that supports educational growth and life-long learning skills. CHS is also committed to producing quality research with distinct societal impact.



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DMS Mission:

The Department of Mathematics and Sciences is committed to offering high-quality education in the mathematics and sciences field, to enhance the students' logical thinking and analytical skills. The department supports the research in different scientific disciplines with distinct societal and scientific impacts.

I. Course Description:

- II. In this course, students will review and learn mathematical techniques necessary for success as an engineer, both in future coursework and on the job. Given the difficulty of solving more realistic engineering problems analytically, the emphasis will be on the understanding and use of computational algorithms. In the process, students will develop a strong working knowledge of software packages like Matlab, Maple, and Mathematica, which are integrated technical computing environments that combine numeric computation, advanced graphics and visualization, and a high-level programming language.

- II. Course Learning Outcomes:** (A summary of intended learning outcomes of the course in each domain of learning). On the successful completion of this course, students will be able to demonstrate the following:

Skills	Course Learning Outcomes
Knowledge & Understanding	1.1 Use different algorithms to find the roots of nonlinear functions and analyze the error
Skills	2.1 Implementation of numerical algorithms applied to interpolation techniques. 2.2 Use numerical algorithms to compute differentiation and integration.
Values	3.1 Use numerical techniques to solve initial value problems. 3.2 Demonstrate skills on computers using MATLAB or other software packages to solve numerical problems. 3.3 Compute solution of system of linear equations using numerical algorithms



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III. Tentative Weekly Course Schedule: (Should mention the specific course topics to be covered within the semester) *May change to accommodate guest presenters & student needs.*

WEEK NUMBER	DATE	UNIT/ TOPIC	CLO(s) alignment	STUDENT TASKS (assignments, readings, exams, quizzes)	No. of Contact hours
1	Dec 21-25	1.1 Review of calculus 1.2 Round-off Errors and Computer Arithmetic	CLO 1		4
2	Dec 28 – Jan 1	1.3 Algorithms and Convergence 2.1 The Bisection Method	CLO 1		4
3	Jan 4-8	2.2 Fixed point Iteration 2.3 Newton's Method	CLO 1	Quiz 1	4
4	Jan 11 – 15	2.4 Error Analytic for Iteration Methods	CLO 1 CLO 2		4
5	Jan 18-22	2.5 Accelerating Convergence	CLO 1, CLO 2		4
6	Jan 25 - 29	3.1 Interpolation and Lagrange Polynomials 3.2 Data Approximations	CLO 2	Quiz 2	4
7	Feb 1-5	3.3 Divided Differences 3.4 Hermite Interpolation	CLO 2		4
Major I, Monday, Feb 9, 2026					
8	Feb 8-12	3.5 Spline Interpolation	CLO 2	)	4
9	Feb 15-18	4.1 Numerical differentiation 4.3 Elements of Numerical Integration	CLO 3		4



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Ramadan and Eid Holiday from Feb 18- March 28					
10	March 29- April 2	4.4 Composite Numerical Integration	CLO 3		4
11	April 5-9	5.1 Elementary Theory of Initial Value Problems 5.2 Euler 's Method	CLO 4	Quiz 3	4
12	April 12 - 16	5.3 Taylor's Methods 6.1 Linear system of Equations	CLO 4		4
Major II, Monday, April 20 th , 2026					
13	April 19 - 23	6.2 Pivoting Strategies 6.3 Linear Algebra and Matrix Inversion	CLO 4 CLO 5		4
14	April 26 - 30	7.1 Norms of Vectors and Matrices 7.2 Eigenvalues and Eigenvectors	CLO 5 CLO6	Quiz 4	4
15	May 3 - 7	7.3 Jacobi and Gauss-Seidel Iterative techniques	CLO 5 CLO6		4
Thursday, May 7: Last Day of Classes					
16	May 9 - 21	Final Exam		Wednesday, May 13 th at 8:30 am	

- IV. Student Assessment & Teaching Strategies: Assessment Task** (Indicate the kind of assessment tasks to be used to measure student learning in each of the learning domains. Example: quiz, oral examination, group work, etc.).
- Teaching Strategies:** (Please indicate the teaching and student activities to be used to develop the kinds of learning involved in each learning domain. Also, research specialized Information about Best Teaching Practices for the particular course/field).

Domain	Assessment Task	Teaching Strategy
Knowledge	Direct Assessment Tool Quiz, Exams Assignments	Lectures using slides and explaining on board.



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	Indirect Assessment Tool: Course Exit Survey	
Skills	Direct Assessment Tool Quiz, Exams Assignments Indirect Assessment Tool: Course Exit Survey	Lectures; Problem Solving sessions; Hands-on Practice; Examples presented in class; Mapping all the limit and differentiation techniques
Values	Direct Assessment Tool Quiz, Exams Assignments Indirect Assessment Tool: Course Exit Survey	Lectures Tutorials Problem Solving sessions Hands-on Practice Solve by MATLAB or Excel, Group work(Project)

V. Course Requirements [Whatever tasks and assignments you include in your course should be aligned with the specified learning outcomes (final learning, skills, knowledge, attitudes, and values the students leave the course with) you have defined and specified earlier.

These requirements should be consistent with the Course Specification on file in the particular department.] **Online activities and exercises:** There will be regular activities during class scheduled that will count towards your grade at the end of the semester.

Midterm: There are two midterm assessments (Major 1 and Major 2) to assess both your understanding of the theory and the practical application of it in the field of Numerical Analysis.

Project: A group project will be assigned to evaluate the understanding of a particular chapter.

Final Exam: The final exam is comprehensive and will assess both your understanding of the theory and the practical application of it in the field of evaluation and assessment for the topics covered throughout the semester.

VI. Schedule of Assessment (Specify the schedule/date of assessment & proportion of assessment, attach related rubrics for each assessment if applicable. For examples of rubrics, visit www.irubrics.com . The schedule should be consistent with the academic calendar.



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Any significant changes should be avoided. If significant changes are made, academic leaders should be informed in writing, and students should be given a new schedule of assessment table).

Assessment	Assessment Task	Week Due	Proportion of Final Assessment
1	First Major Exam	Week 7: (Ch. 1, Ch. 2)	20%
2	Second Major Exam	Week 12: (Ch 3, Ch 4)	20 %
3	4 Quizzes	Weeks: 4, 6, 11, 14	10%
4	6 Homework Assignments	Weeks: 2, 5, 8, 9, 10, 13	5%
6	Project	13 th Week	5%
7	Final Exam		40%
8	Total		100%

VII. Learning Resources

- A. References - Textbook:** R. L. Burden and J. D. Faires, Numerical Analysis, Brooks/Cole, 10th ed., 2011. **Notes Website:** Holistic Numerical Methods at <http://mathforcollege.com/nm/>
- B. Facilities Required** – (online – laptop/internet, lecture room, computing resources, laboratories, others)
- C. EDU-HUB Platform** – The platform will be used for all online education experiences at PSU. It supports a variety of methods to support the teaching and learning strategies across all disciplines. It also includes the LMS.

VIII. Classroom Policies

- A. Academic Integrity Policy (e.g., plagiarism or dishonesty)**



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“Plagiarism can be defined as unintentionally or deliberately using another person’s writing or ideas as though they are one’s own. Plagiarism includes, but is not limited to, copying another individual’s work and taking credit for it, paraphrasing information from a source without proper documentation, and mixing one’s own words with those of another author without attribution. In addition, buying a paper or project, or downloading a paper from the Internet, and submitting them as your own is also plagiarism. The penalty for academic dishonesty will bring course expulsion and failure, or even suspension” (Academic Integrity and Syllabus Acknowledgement Form).

All students are expected to submit their ‘own’ work and not the work of others.

B. Attendance Policy

Please, adhere to the following guidelines:

1. The University attendance policy will be strictly followed. In this course, the absence of **(insert number here based on the number of credit hours)** hours results in a Denied Notice (DN). (Refer to the student regulation handbook for further details.)

Please Note: IT IS THE SOLE RESPONSIBILITY OF THE STUDENT TO KEEP SATISFYING their level of attendance. Otherwise, a “DN” grade will be granted automatically during any time in the semester without any notice from the e-register system.

2. No make-up exams will be allowed except for university accepted documents. (Refer to the student regulation handbook for further details.)

C. Homework Submission Policy *(policy depends upon the course or the instructor)*

Your instructor will determine a deadline for submitting each homework. Five points will be assigned to the Homework. The due dates are announced in the assessment table. It is the sole responsibility of the student to ensure that he/she has completed the homework before the stated deadlines. Please note that after the deadlines have passed, homework, quizzes, will not be graded for any reason.

IX. My Assumptions (Optional)

Tips on how to enhance your problem-solving abilities:

- Attend all classes and participate in the class.



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- Do all the homework assignments on time. It is impossible to understand math without solving problems.
- You are urged to practice (but not memorize) more problems than the ones assigned.
- You should always try to solve a problem on your own before reading the solution or asking for help.
- If you find it difficult to handle a certain type of problems, you should try more problems of that type.
- You are encouraged to solve some of the review problems at the end of each chapter.
- The practice you get doing homework and reviewing the class lectures will make exam problems easier and quicker to solve.
- Many videos are provided from the instructors of the course watch it and contact your instructor via email, if you have any concern.
- Many problems will be solved on MATLAB / excel, so students will have a good knowledge of different mathematical programs.

Tips on how to be successful in Numerical Analysis (Math 221)

It is expected that you should:

1. Attend all classes, pay attention, and do not hesitate to ask questions;
2. Spend about 4 hours each week working homework suggested problems, reviewing the class lectures, reading the text book, and studying for quizzes and exams;
3. Try to solve problems on your own before reading solutions or asking for help.
4. Consult your instructor immediately when you face a continuous difficulty in the course;
5. Complete all homework suggested problems every week, check the correctness of your work and understand the methods and principles they illustrated;
6. Master the designed course topics before each quiz and exam, and if necessary, complete additional problems beyond those assigned and consult other sources if you find the assigned problems and text are insufficient;
7. Write your solutions to homework, exams, and quizzes problems in an organized way;
8. Have good knowledge of MATLAB.